Super Admin ("The Boss") Core Module: `ADM` (Platform Administration & Tenancy Control)

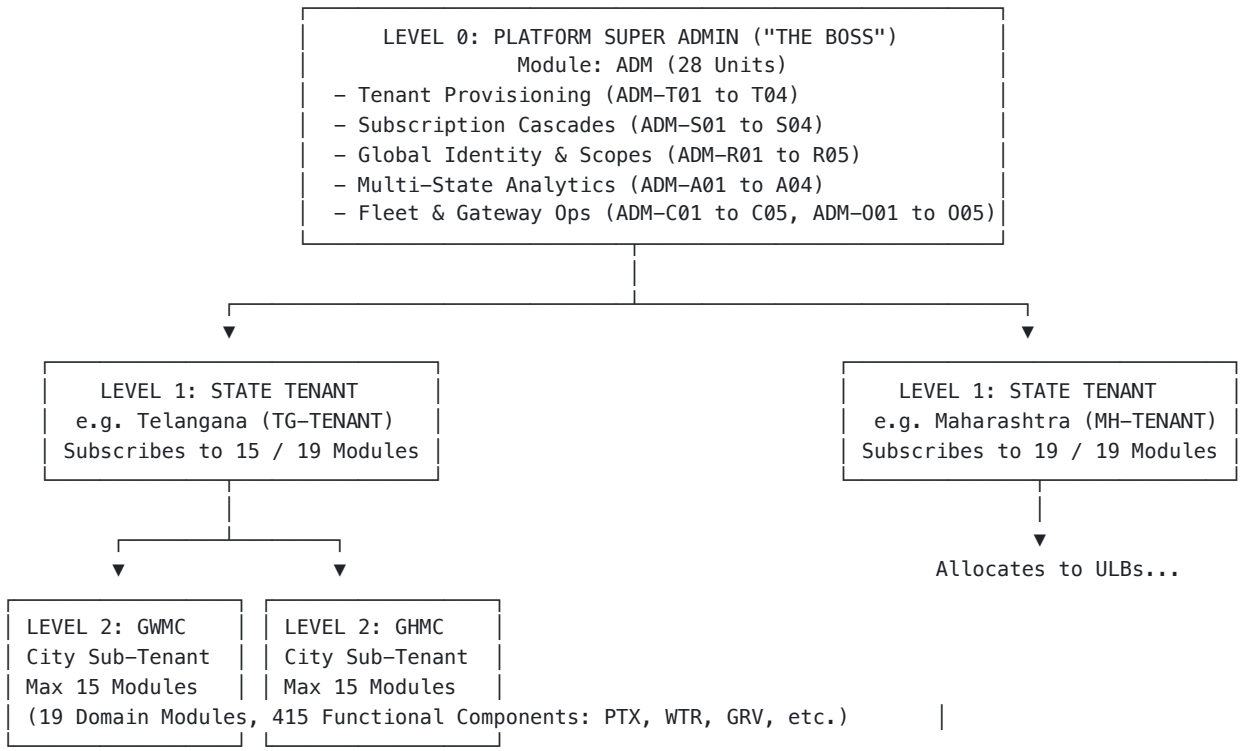
Super Admin operates at **Level 0 (Global SaaS Control Plane)** across all States and Cities. Just as the 19 municipal domain modules have 415 components, the Platform Administration module (`ADM / CTL`) requires **28 granular components** organised into **6 functional domains**:

Component Matrix: 28 Super Admin Control Units

Domain	Comp Code	Component / Process Name	Operational Scope & Power
1. Multi-Tier Tenancy & Provisioning	`ADM-T01`	**State Tenant Provisioning**	Onboard new State (e.g., Telangana `TG`, Maharashtra `MH`), configure state codes, timezones, official languages.
	`ADM-T02`	**Sub-Tenant (ULB / City) Provisioning**	Provision child municipalities (e.g. GWMC, GHMC) under State Tenants, define ward/zone boundaries and municipal grades.
	`ADM-T03`	**Tenant Lifecycle & Status Controls**	Suspend, lock, archive, or decommission tenant workspaces; maintenance mode switch per tenant.
	`ADM-T04`	**Tenant Domain & White-Labelling**	Custom subdomain / CNAME routing (e.g. `gwmc.urbanerp.gov.in`), state logos, watermarks, official seal management.
2. Module Subscription & Packaging	`ADM-S01`	**Global 19-Module Master Catalogue**	Activate/deactivate platform modules, define module feature flags, versioning, and deprecation policies.
	`ADM-S02`	**Tenant Subscription Allocator**	Assign module subscriptions to State Tenants (e.g., State A subscribes to 12 modules; State B to all 19).
	`ADM-S03`	**Sub-Tenant Cascade Validator**	Set allocation ceilings for Sub-Tenants (ensures Cities cannot activate modules unallocated by their parent State).
	`ADM-S04`	**Quota & Usage Tier Management**	Allocate storage quotas, API call limits, concurrent user caps, and SMS/WhatsApp notification credits per tenant.
3. Global RBAC & Identity Control	`ADM-R01`	**Global User Administration**	Provision and manage Level 0 (Super Admin), Level 1 (State Admin), and Level 2 (City Admin) user identities.
	`ADM-R02`	**Cross-Tenant Role & Scope Matrix**	Assign multi-membership scopes (e.g., state auditor accessing 5 cities; regional commissioner overseeing 3 ULBs).
	`ADM-R03`	**Component RBAC Permission Engine**	Manage the 415 granular action definitions (`{MODULE}-{ID}:{ACTION}`); build custom permission bundles.
	`ADM-R04`	**Session & Impersonation Hub**	Secure administrative impersonation ("Log in as tenant officer" with strict read-only / audit logging for support).
	`ADM-R05`	**Universal SSO & Identity Brokering**	Keycloak realm federation, State SSO bridge (e.g. MeriPehchan, Aadhaar e-Pramaan), MFA & hardware key enforcement.
4. Executive Cross-State Analytics ("The Boss View")	`ADM-A01`	**Pan-State Macro Dashboard**	Real-time consolidated KPIs: total revenue collected, active properties, live grievances across all states and cities.

Domain	Comp Code	Component / Process Name	Operational Scope & Power
	ADM-A02	**State & City Comparative Benchmarking**	Side-by-side performance indices: DCB collection efficiency, grievance SLA adherence, Swachh Survekshan scores.
	ADM-A03	**Cross-Tenant Financial Consolidation**	Consolidated municipal balance sheets, central grant utilisation (15th Finance Commission, AMRUT, SBM) across states.
	ADM-A04	**Spatial GIS Operations Command**	Multi-city PostGIS layer viewer: view parcels, pipelines, and works across multiple ULB jurisdictions in one map.
5. Integration, Security & Compliance	ADM-C01	**Central Gateway & Rate Limiting**	Kong / APISIX gateway policies, per-tenant API rate limits, IP whitelisting, and Web Application Firewall (WAF) rules.
	ADM-C02	**Payment Gateway Hub (Multi-Aggregator)**	Provision master merchant accounts (SBI ePay, BillDesk, Razorpay, BBPS central switch) and assign to tenant escrow accounts.
	ADM-C03	**State Adapter Gateway**	Central state integration hubs: e-Courts CNR sync, IGRS registration sync, CGF/Treasury sync, and DISCOM feeds.
	ADM-C04	**Immutable Audit Vault (CERT-In Compliance)**	Central security logs, tamper-evident audit trails with cryptographic hash chaining (who accessed what, when, across all tenants).
	ADM-C05	**Data Sovereignty & RLS Verification**	PostgreSQL Row-Level Security policy validation, tenant data leakage detection, and tenant-isolated backup triggers.
6. System Health, Fleet & Upgrades	ADM-O01	**Fleet Health & Microservice Topology**	Live health status of all 19 module service pods, database connections, Redis clusters, and Flowable BPMN engines.
	ADM-O02	**Rolling Schema Migration Orchestrator**	Run zero-downtime database migrations (Alembic) across all tenant database schemas / tenant IDs.
	ADM-O03	**Cache & Redis Scoped Eviction**	Invalidate and purge scoped cache partitions (`stats:tenant:{id}:*`) without flushing global cluster cache.
	ADM-O04	**Platform Backup & Disaster Recovery**	Automated snapshot orchestration, Point-In-Time-Recovery (PITR) drill triggers, and cross-region failover.
	ADM-O05	**Global Announcement & Broadcast Banner**	Send platform-wide or per-state emergency notifications, scheduled maintenance alerts, and advisory broadcasts.

Architectural Placement



This ensures full administrative separation: Super Admin manages the entire SaaS engine, subscriptions, and cross-state analytics without being restricted by individual city or state boundaries.